Introduction to Machine Learning
Kaggle Challenge
Due Date for Submission: As per guidelines in Kaggle
Due Date for Report: 2 Days after Kaggle Deadline

Kaggle Competition Link:
https://www.kaggle.com/t/83081d62f0604355948c9313230de90d

Goal

Predict whether a patient is readmitted to the hospital within 30 days of discharge (readmit_30d) using the multi-table hospital dataset. Submissions are scored with ROC-AUC.

Dataset

The data comes as many linked tables, not a single file. Most of your work is turning it into a clean, one-row-per-encounter dataset, so the quality of your data pipeline matters as much as your choice of model.

Tasks

1. Build your modeling table. Join and aggregate the provided tables into one row per encounter_id, for both the train and test encounters. You do not have to use every table. Decide which tables are useful and discard the rest.
2. Preprocess. Inspect and clean the raw data (formats, units, duplicates, implausible values), handle missing values, and encode categorical variables. Fit any imputer, scaler or encoder on training data only.
3. Train and compare models. Try different models from the course (for example Naive Bayes, KNN, Decision Tree, Random Forest, AdaBoost, etc).
4. Submit to Kaggle. Submit predicted probabilities (not 0/1 labels) for every encounter_id in test.csv, in the format of sample_submission.csv.

Deliverables

A. Report

The report must cover:

1. Data pipeline. Which tables you used and which you dropped (and why), how you joined them, and how you aggregated each table that has many rows per encounter (for example last value, mean, count).
2. Final dataset. The number of rows and columns, and a list of every column with its source table and how you created it. You may attach this as a file named final_columns.csv or put it in an appendix.
3. Data quality. Problems you found in the raw data and how you handled them.
4. Missing values. Where values were missing, what you think caused it, and what you did about it.

5. Models. A comparison table of at least your best three models with their key hyperparameters, how you tuned and evaluated them, and the score each achieved.
6. Results and discussion. Your final public leaderboard score, which model performed best and why, and what you would try next.

B. Code

An organized notebook that runs from the raw CSV files to your final submission file.

Rules

- Use only the data provided in the competition.
- You are responsible for being able to explain every step of your submission. You may be asked to walk through your pipeline.
- Do not share code or predictions privately with other students.

C. Rubrics

Marks	Assessment Criteria
8	Top 10% on the Kaggle leaderboard. All tasks are successfully completed, and multiple attempts are observed on leaderboard. Code is well-structured, readable, and properly commented. Instructions are carefully followed, and the solution demonstrates exceptional effort, sound ML practices, and performance beyond the minimum requirements
7 – 8	Top 50% on the Kaggle leaderboard. All tasks are successfully completed, and multiple attempts are observed on leaderboard. Code is well-structured and properly commented. Instructions are followed, and the solution exceeds the minimum requirements with good ML practices
5 - 7	Good performance on the Kaggle leaderboard. Tasks are successfully completed. Code is readable and well-written. Instructions are followed, and the solution meets the minimum requirements
3 - 5	Moderate performance on the Kaggle leaderboard. Most tasks are completed, but the solution has some limitations in predictive performance, implementation, or code quality. Basic instructions are followed, although some improvements are needed in model selection, evaluation, or documentation
0 – 3	Below expectations. Tasks are incomplete or the solution demonstrates significant limitations in predictive performance, implementation, or code quality. Instructions may not be fully followed, and the submission does not adequately meet the minimum requirements